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Identifying Barriers to Human Papilloma Virus (HPV) Vaccination in US Veteran Population

Jose Zevallos, Principal Investigator

Co- Investigator: Angela L. Mazul, Kelly H. Burkitt, Maria Mor

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Summary
Oropharyngeal cancer (OPC), specifically oropharyngeal squamous cell carcinoma, is among the most common cancer of the upper aerodigestive tract and includes cancers arising from the tonsils, base of tongue, soft palate, and lateral and posterior pharyngeal walls. Approximately 60-70% of new OPC cases are now attributable to human papillomavirus (HPV).[1] HPV (+) OPC incidence has continuously risen since the 1980s in the United States, while cancer resulting from more traditional risk factors such as smoking and tobacco have decreased.[2] In the last 10 years, HPV (+) OPC has overtaken cervical cancer as the most common HPV-related malignancy in the United States.[3] Oral oncogenic infection is a necessary but insufficient precursor to HPV (+) OPC.[4] Previous studies have found that among unvaccinated men aged 18 to 59, the prevalence of oncogenic and non-oncogenic oral HPV infection was 1.6% and 8.4%, respectively.[5]

While the vast majority of people clear the infection, a small subset of individuals develop chronic HPV infection that can lead to HPV (+) OPC.[6,7] Heavy smokers also have nearly four times the prevalence of oral HPV compared with non-smokers. Smoking has been established as a negative prognosticator in this disease process.[8] However, HPV (+) tumors respond more effectively to radiation and chemotherapy than their HPV (-) counterparts.[9-11] Therefore, current treatment paradigms and clinical trials for OPC have focused on de-intensifying therapy and minimally invasive surgical approaches to limit treatment morbidity and improve functional outcomes.[12-15] While HPV-positive tumors of the oropharynx have a better prognosis than their HPV-negative counterparts, these and other HPV-related malignancies are preventable with HPV vaccination. Notably, oropharyngeal cancer incidence is rising significantly at the VA among various racial groups and all age groups. When compared to the US general civilian population, HPV vaccination rates in active duty military and Veteran populations are extremely low.[16] In contrast, HPV vaccination rates doubled between 2013 and 2018 in the civilian population.[17] The Center for Disease Control recently expanded HPV vaccination indications to include men and women <45 years of age, making many Veterans entering the Veteran's Health Administration (VHA) system eligible.[18]

Using data from the VHA Corporate Data Warehouse (CDW), we will estimate the prevalence of HPV vaccination in Veterans <45 years of age and examine vaccination rates by race/ethnicity, socioeconomic status, smoking status, rurality based on Rural-Urban Commuting Area Codes (RUCAs), and geographical location based on Veteran Integrated Service Network (VISN). VISNs are responsible for VHA healthcare planning and resource allocation in a particular geographic region,[24] and thus present an ideal opportunity to assess geographic variability in vaccination rates and ultimately apply targeted interventions. We hypothesize that smokers and African Americans have the lowest rate of HPV vaccination, and HPV vaccination rates will vary significantly across regions, with rural, South-central, and Southeast regions demonstrating the lowest rates of HPV vaccination. This approach will identify at-risk groups based on demographic or regional characteristics and allow for tailored VHA or VISN-specific HPV vaccination campaigns or other cancer prevention strategies, including smoking cessation.

List of Abbreviations

Corporate Data Warehouse (CDW)

Executive Leadership Team (ELT)

Human papillomavirus (HPV)

Joint Legacy Viewer (JLV)

Oropharyngeal Cancer (OPC)

Personal Health Information (PHI)

Record Control Schedule (RCS)

Rural-Urban Commuting Area Codes (RUCAs)

Veteran Integrated Service Network (VISN)

Federal Information Processing System (FIPS)

Planning Systems Support Group (PSSG

Contents

1.0 Study Personnel

Jose Zevallos is the Chairman of the Department of Otolaryngology at UPMC. He was previously Division Chief of Head and Neck Surgery and the Joseph B. Kimbrough Endowed Professor of Head and Neck Surgery in the Department of Otolaryngology/Head and Neck Surgery at Washington University School of Medicine, St. Louis, MO. He also served as program director of the advanced head and neck surgical oncology and microvascular reconstructive surgery fellowship. He recently was also asked to serve as the interim Deputy Director of the Hillman Cancer Center.

Angela L. Mazul, PhD, MPH is an Assistant Professor in the Department of Otolaryngology at the University of Pittsburgh and a member of UPMC Hillman. Dr. Mazul is a cancer epidemiologist specialing in health and neck and HPV related cancers. In addition, she will focus on understanding and addressing health disparities in cancer presentation and outcomes. Her research will develop a robust head and neck cancer population-based study.

Kelly H. Burkitt, PhD is a clinical health psychologist with 30+ years of health-related research experience and particular expertise in program and quality improvement (QI) evaluation. In addition to serving as a co-investigator on numerous VA Health Systems Research (HSR; formerly Health Services Research & Development) awards, she guided many VHA QI evaluation projects for VA Pittsburgh Healthcare System (VAPHS), VA Office of Analytics and Performance Integration, Office of Health Equity (OHE), and Office of Patient Care Services (PCS). Dr. Burkitt is well-versed in local and national human subjects and data security policies, having served on the VAPHS Institutional Review Board (IRB) and Research and Development (R&D) Committees. She is also a founding member of the HSR Veteran Engagement Workgroup and chaired its subgroup on Veteran engagement at the COIN level.

Maria Mor, PhD, is the Pittsburgh Director of the Biostatistics and Informatics Core at the Center for Health Equity Research and Promotion. She is a biostatistician with 20 years of experience conducting research and quality improvement projects in VA. She has collaborated on many projects assessing quality, safety, outcomes, for a variety of Veterans populations with a focus on those with vulnerabilities such as due to historically disadvantaged groups (race, gender) or comorbidity burden (chronic disease). She is familiar with a wide variety of data sources available for research within VA and adept at working with a guiding programmers and analysts within the BIC in extracting and analyzing data.

- **Collaborators/Other Institutions -** Not Applicable

- **Non-Financial Conflicts of Interest -** None

Package uploads (Section 1.0):
- **Research Staff Form**

2.0 Introduction

Background:

Oropharyngeal cancer (OPC), specifically oropharyngeal squamous cell carcinoma, is among the most common cancer of the upper aerodigestive tract and includes cancers arising from the tonsils, base of tongue, soft palate, and lateral and posterior pharyngeal walls. Approximately 60-70% of new OPC cases are now attributable to human papillomavirus (HPV).[1] HPV (+) OPC incidence has continuously risen since the 1980s in the United States, while cancer resulting from more traditional risk factors such as smoking and tobacco have decreased.[2] In the last 10 years, HPV (+) OPC has overtaken cervical cancer as the most common HPV-related malignancy in the United States.[3] Oral oncogenic infection is a necessary but insufficient precursor to HPV (+) OPC.[4] Previous studies have found that among unvaccinated men aged 18 to 59, the prevalence of oncogenic and non-oncogenic oral HPV infection was 1.6% and 8.4%, respectively.[5]

While the vast majority of people clear the infection, a small subset of individuals develop chronic HPV infection that can lead to HPV (+) OPC.[6,7] Heavy smokers also have nearly four times the prevalence of oral HPV compared with non-smokers, and smoking has been established as a negative prognosticator in this disease process.[8] However, HPV (+) tumors respond more effectively to radiation and chemotherapy compared to their HPV (-) counterparts.[9-11] Therefore, current treatment paradigms and clinical trials for OPC have focused on de-intensifying therapy and minimally invasive surgical approaches to limit treatment morbidity and improve functional outcomes.[12-15] While HPV-positive tumors of the oropharynx have a better prognosis than their HPV-negative counterparts, these and other HPV-related malignancies are preventable with HPV vaccination. Notably, oropharyngeal cancer incidence is rising significantly at the VA among various racial groups and all age groups. When compared to the US general civilian population, HPV vaccination rates in active duty military and Veteran populations are extremely low.[16] In contrast, HPV vaccination rates doubled between 2013 and 2018 in the civilian population.[17] The Center for Disease Control recently expanded HPV vaccination indications to include men and women <45 years of age, making many Veterans entering the Veteran's Health Administration (VHA) system eligible.[18]

3.0 Objectives

- **Study Objectives**

Estimate the prevalence of HPV vaccination in Veterans less than <45 years of age and identify Veteran populations at highest future risk for HPV (+) OPC by identifying barriers to HPV vaccination.

Using data from the VHA Corporate Data Warehouse (CDW), we will estimate the prevalence of HPV vaccination in Veterans <45 years of age and examine vaccination rates by race/ethnicity, socioeconomic status, smoking status, rurality based on Rural-Urban Commuting Area Codes (RUCAs), and geographical location based on Veteran Integrated Service Network (VISN). VISNs are responsible for VHA healthcare planning and resource allocation in a particular geographic region and thus present an ideal opportunity to assess geographic variability in vaccination rates and ultimately apply targeted interventions. We hypothesize that smokers and African-Americans have the lowest rate of HPV-vaccination, and HPV vaccination rates will vary significantly across regions, with rural, South-central, and Southeast regions demonstrating the lowest rates of HPV vaccination. This approach will identify at-risk groups based on demographic or regional characteristics and allow for tailored VHA or VISN-specific HPV vaccination campaigns or other cancer prevention strategies including smoking cessation.

- **Hypothesis**

  We expect that HPV vaccination is lower among African-American Veterans compared to non-Hispanic Whites, as has been noted in the civilian population.[17] The VHA Office of Rural Health has determined that Veterans living in rural communities often lack adequate access to basic healthcare and preventative care,[33] and we hypothesize that HPV vaccination rates will be lower among Veterans living in rural communities. We also expect Veterans in south-central and southeast VISNs to have the lowest rate of HPV vaccination. This hypothesis is derived from civilian HPV vaccination data among adolescents, where these regions have among the lowest vaccination rates in the nation. While data from civilian populations serve as hypothesis-generating, the lack of data on barriers to vaccination within the VHA severely limits the potential for tailored prevention programs. With at least 1.7 million samples (including 283,000 females) in the VHA <45 years of age, we will be adequately powered to detect any association. However, with this large number of samples, there is the risk that every test will be statistically significant, especially when comparing the prevalence of HPV vaccination across variables. Thus, we will focus a priori on effect sizes and differences in prevalence rather than p-values.

- **Relevance to Veterans**

  While HPV vaccination guidelines have been expanded to include a larger age range of patients, there is currently no requirement for HPV vaccination among active-duty military or Veterans.[16,19] Recent data demonstrated that 19% of males in active duty developed a new HPV16 or HPV18 oncogenic infection over ten years of service.[20] Veterans also smoke at a higher rate compared to the general population[21,22] and the 2019 Survey of Veteran Enrollees' Health and

Use of Health Care showed no change in the rate of smoking as well as a high rate of unsuccessful cessation attempts.[23] Together, low HPV-vaccination and high smoking rates among this population make US Veterans more susceptible to oropharyngeal cancer and other HPV-related malignancies. We believe that identifying factors associated with the low HPV-vaccination rates among Veterans will be the first step to reducing future incidence of HPV-related malignancies such as OPC and alleviate suffering in the VHA population. Our proposed study would be the first to examine barriers to HPV vaccination in the Veteran population.

4.0 Resources

- **VAPHS Resources**

  Dr. Zevallos will have an office in CHERP located within the Research Office Building, 2nd floor, Suite 2A126. Dr. Burkitt's and Dr. Mor's offices are also in the Research Office Building CHERP's Biostatistics and Informatics Core staff will compile and analyze the data.

- **Service Line Impact**

  There will be no service line impact for this study.

- **Participant Payments**

  None

- **Off-Site Ancillary Activities**

  N/A

Package Uploads (Section 4.0): - Not applicable
- **Letters of Support**
- **Agreements**
- **Budget**

5.0 Study Procedures

5.1 Study Design

Study Population: We will construct a cross-sectional, national sample of Veterans using data from the VHA CDW to calculate the prevalence of HPV vaccination.

Inclusion and Exclusion Criteria: To ensure that we capture a sample that qualifies for HPV vaccination and engages with the VHA and military health care system, we will include all Veterans with at least one primary care visit from January 2018 to

June 2024 aged <45 years. In 2017, 6.1 million Veterans had one VA visit, of which 1.7 million were aged <45 years.[25]

Preliminary Data. In a feasibility study using the CDW to query HPV vaccination data and conducted at the St. Louis VA, we identified 1,442,908 Veterans aged < 45 years seeking care within the VHA from 1 January 2018 to 31 December 2019. We then identified 54,736 Veterans with a history of HPV vaccination.

Data Collection: The primary variables of interest in this study are HPV vaccination status/rates by race/ethnicity, rurality, VISN, smoking status, gender, and age. Other variables necessary for robust examination of the research question include patient diagnoses (e.g., HPV-related malignancies or comorbidities), other sociodemographic variables, branch of service, alcohol use, and healthcare utilization dates. Study data will be abstracted from the VHA CDW, VistA, quarterly Area Deprivation Index within VINCI. All personal health information (PHI) and datasets will be maintained exclusively on VINCI servers or the VAPHS research server, behind the VA firewall. The server is located within the Heinz Division at the Veterans Affairs Pittsburgh Healthcare System (VAPHS). It is accessible only through the VA system.

HPV Vaccination: Vaccination records are available in the CDW for the study population from which we will abstract the date of each HPV vaccination dose. HPV vaccination recommendations include a series of 3 doses at minimum intervals of 4 weeks between the first and second dose, 12 weeks between the second and third dose, and five months between the first and third dose.[26] Therefore, we will classify HPV vaccination into multiple categories that account for the initiation of vaccination and compliance as outcome variables: (1) any HPV vaccination (yes/no), (2) number of HPV vaccines (0-3), (3) completion of 3-dose HPV vaccination recommendations (yes/no), and (4) HPV vaccine time as routine (yes/no) if vaccinated between age 9 through 12 years, catch-up as 13 to 26 years, and shared-decision making as vaccinated between ages 26 to 45 years.

Study Procedures-Smoking, race/ethnicity, rurality, geographical location, and age as potential risk factors: Smoking status will be derived from automated clinical reminders known as Health Factors in VHA CDW.[27,28] The VHA requires providers to complete these Health Factors regularly within the electronic health records. The results from these structured EHR data are housed in the CDW and refreshed daily. Smoking status will be determined using a VHA-specific validated structured data tool.[27,28] We will identify more recent smoking-related Health Factors and apply the same methodology to categorize and use the Heath Factors for this study. We will classify smoking as former, current, and never on the index date (i.e., primary care visit within the study period) or within one year before the index date. Race/ethnicity will be obtained from VA CDW. Each subject will have an ethnicity category (Hispanic/non-Hispanic). Race will be categorized as white, black, Native American/Indigenous, and Asian/Pacific Islander. We will use the Area Deprivation Index, a neighborhood-level measure of socioeconomic status that is available from

VINCI, as such measures have been previously shown to be associated with outcomes in other cancer types.[30] We will link the Veterans' County of Residence through Federal Information Processing System (FIPS) codes in the Planning Systems Support Group (PSSG) geocoded files to RUCAs. RUCAs were developed by the US Department of Agriculture to delineate rural-urban census tracts, and have been adopted by the VHA Office of Rural Health as the primary stratification system for defining rural/urban status.[31] RUCAs have advantages over other rural-urban measures since they take into account not only population density but also a functional measurement of travel distance. We will classify RUCAs into four categories: urban, large rural, small rural, and isolated. Other risk factors, such as age on the index date will be obtained from CDW.

<u>Validation of HPV-Vaccination Data:</u>

Though HPV vaccination guidelines have been expanded to include patients up to the age of 45, the immunization is most commonly administered to patients in their adolescent, teen, and early adult years. Therefore, we expect that many Veterans with a history of vaccination will have received the immunization before entering the VHA. One study at the Bronx VHA examined HPV vaccination in an institutional cohort of Veterans by querying both the Institutional Data Warehouse as well as medical records through the Computerized Patient Record System (CPRS) and Joint Legacy Viewer (JLV) systems; due to inconsistencies in the recording of vaccinations in various electronic health record systems and a concern that vaccination prior to entry in the VHA (DoD vaccination or prior) may not populate into the vaccination widget of JLV, this study increased confidence in immunization data by individually examining the notes and medical records of patients who were found to have history of HPV-vaccination in the queries.[16]

Per the CDW Immunization factbook, "VistA/CPRS Background Immunization entries were mandated by 10N's Deputy Under Secretary for Health Operations and Management (DUSHOM)". As HPV vaccination was approved for women in 2006 and men in 2009, it appears likely that anyone vaccinated within the VHA or who reported a history of HPV vaccination to their VA primary care provider will be captured in the CDW. After a thorough review of CPRS and JLV, we will compare the HPV vaccination status of Veterans in the CDW against the medical record and quantify the concordance between the administrative data and chart-level data on HPV vaccination. This validation process may also be completed to verify the quality of smoking status information in the VHA's administrative data.

Usual Care or Interventions

- **N/A**

Risks and Benefits

- This retrospective, electronic medical record study carries minimal risk. The primary risk is confidentiality. We will follow all safeguards outlined in this protocol to minimize this risk. Any loss of confidentiality, identifiable information, or research data will be reported to the research office, the facility information security officer, and the privacy officer within one hour of the incident. Only investigators listed on this protocol will have access to data for the entirety of the study.

- While HPV vaccination guidelines have been expanded to include a larger age range of patients, there is currently no requirement for HPV vaccination among active-duty military or Veterans.[16,19] Data demonstrated that 19% of males in active duty developed a new HPV16 or HPV18 oncogenic infection over ten years of service.[20] Veterans also smoke at a higher rate compared to the general population[21,22] and the 2019 Survey of Veteran Enrollees' Health and Use of Health Care showed no change in the rate of smoking as well as a high rate of unsuccessful cessation attempts.[23] Together, low HPV vaccination and high smoking rates among this population make US Veterans more susceptible to oropharyngeal cancer and other HPV-related malignancies. We believe that identifying factors associated with the low HPV vaccination rates among Veterans will be the first step to reducing the future incidence of HPV-related malignancies such as OPC and alleviating suffering in the VHA population. Our proposed study would be the first to examine barriers to HPV vaccination in the Veteran population.
- We believe that identifying factors associated with the low HPV vaccination rates among Veterans will be the first step to reducing the future incidence of HPV-related malignancies such as OPC and alleviating suffering in the VHA population:
- We believe that this proposal will benefit science/society and contribute to generalizable knowledge.
-

Alternatives

- **None**

Package uploads (Section 5.1):

- Study Procedures Table (required for GTM studies and minimal risk studies that use SOC) N/A
- Patient ID Card (if study involves treatment or intervention) N/A
- Screening documents, tests, questionnaires, surveys, or any worksheets participants will complete. N/A
- Case Report Forms None

5.2 Study Population, Recruitment Methods and Informed Consent Process

Study Population: We will construct a cross-sectional, national sample of Veterans to calculate the prevalence of HPV-vaccination with data from the VHA CDW.

Inclusion and Exclusion Criteria

Inclusion and Exclusion Criteria: To ensure that we capture a sample that qualifies for HPV vaccination and engages with the VHA and military health care system, we will include all Veterans with at least one primary care visit from January 2018 to June 2024 aged <45 years. In 2017, 6.1 million Veterans had one VA visit, of which 1.7 million were aged <45 years.[25] Preliminary Data. In a feasibility study using the CDW to query HPV vaccination data, we identified 1,442,908 Veterans aged < 45 years seeking care within the VHA from 1 January 2018 to 31 December 2019. We then identified 54,736 Veterans with a history of HPV vaccination.

Recruitment – N/A

5.3 Data Analysis

Unadjusted analyses: We will examine the prevalence of HPV vaccination compliance, completion, and time frame across the entire cohort, as well as differences in the prevalence or completion of HPV vaccination by smoking status, race/ethnicity, VISN, or rural-urban context. Chi-square tests will be used to determine if there are differences in the prevalence of HPV vaccination between smoking status strata, race/ethnicity, VISN, or rural-urban context. We will also calculate prevalence differences to measure the potential public health impact of the risk factor by calculating the number of additional Veterans vaccinated. Adjusted analyses: We will use logistic regression for outcomes (1), modeling no (vs any) HPV vaccination; for outcome (2) modeling no completion (vs completion) of HPV vaccination recommendations; and for outcome (4) modeling no routine (vs routine) HPV vaccination. For outcome (2), we will use multinomial logistic regression for HPV dose (0, 1 or 2, and 3). To identify barriers to HPV vaccination, we will include all of the aforementioned candidate risk factors as independent variables in the regression analyses. If any geospatial differences in the prevalence of HPV vaccination are identified in the unadjusted analyses, we will consider the innate relationships of individuals within a neighborhood with multilevel (multinomial) logistic regression, as appropriate.[32] Additionally, we will study the interaction between these risk factors and VISNs to determine the best methods for targeting intervention.

Validation of HPV-vaccination data: Concordance between HPV-vaccination status (yes vs no and number of doses) in CDW and chart-level data for a subset of Veterans in the national cohort who were seen for a primary care visit in the study period within Pittsburgh VHA system will be quantified by determining the sensitivity and specificity of the HPV-immunization data in the CDW.

5.4 Withdrawal of Subjects – N/A

6.0 Data Safety Monitoring – N/A

7.0 Privacy and Confidentiality

In this study, all personal health information (PHI) and datasets will be maintained exclusively on VINCI servers or the VAPHS research server, behind the VA firewall. The server is located within the Office of Information & Technology Data Center at the Veterans Affairs Pittsburgh Healthcare System. It is accessible only through networked computers. Only project staff listed within this protocol will have access to the data. If study personnel are no longer part of the research team, access to research study data will be removed on their last day of working on the study.

- The data collected for this study will guide future studies in this area of research and be utilized in an IIR proposal.

Package Uploads (Section 7.0):

- **VHA Research Protocol Privacy Review Checklist (VA Form 10-250)**
- **Department of Veterans Affairs Enterprise Research Data Security Plan ERDSP in the submission package.**
- **DMAP or Resource Sharing Plan if this research is part of a grant.**
- **VA Data Storage and Retrieval Worksheet if you will be collecting and/or storing VA research data at a non-VA location**
- **Honest Broker Certification, if applicable**

8.0 Collaborative/Multisite Research – N/A

9.0 References

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